QML: FINAL PROJECT

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Abstract

A classical Generative Adversarial Network (GAN) consists of two neural networks—the generator and the discriminator—trained in competition. The generator maps samples from a latent noise distribution into synthetic data that aims to fool the discriminator, while the discriminator learns to distinguish real from generated data. Through adversarial training, both models are optimized toward an equilibrium.

A quantum GAN (QGAN), as introduced in paper (4) [1], implements this framework within a hybrid quantum-classical architecture. A quantum encoder embeds classical randomness into quantum states, and a variational quantum circuit (parameterized quantum circuit) acts as the generator. Generated samples arise from measuring operator expectation values, while a discriminator—either classical or quantum—guides the learning process. Optimization leverages automatic differentiation tools to train the variational parameters.

The potential advantage of a QGAN lies in the representational power of quantum systems. Quantum superposition and entanglement allow compact encoding of high-dimensional probability distributions, which may enable learning richer distributions than classical GANs and potentially provide efficiency or expressivity advantages.

1 Introduction: Classical and Quantum GANs

Generative modeling has become a central theme in modern machine learning, enabling the creation of models that learn complex probability distributions and generate realistic synthetic data. Among the most influential frameworks is the Generative Adversarial Network (GAN), introduced by Goodfellow et al. [2]. In a GAN, two neural networks—a generator and a discriminator—engage in an adversarial game: the generator attempts to produce synthetic data that resembles the real dataset, while the discriminator learns to distinguish between genuine and generated samples. This

adversarial training paradigm has led to significant progress in image synthesis, super-resolution, domain translation, and data augmentation [3, 4, 5].

Parallel to these developments, quantum machine learning (QML) has emerged as a promising field that seeks to combine quantum computation with machine learning algorithms [6]. By leveraging quantum mechanical phenomena such as superposition and entanglement, QML algorithms may offer computational advantages over their classical counterparts in terms of representational efficiency and training complexity. Within this broader context, quantum analogues of GANs have been proposed as a way to extend the powerful generative framework into the quantum domain.

One notable approach is the Variational Quantum Generator (VQG), introduced by Romero and Aspuru-Guzik [1]. The VQG adapts the classical GAN architecture into a hybrid quantum-classical system, where the generator is implemented as a variational quantum circuit that encodes classical randomness into quantum states and transforms them via parameterized unitaries. Samples are obtained by measuring observables, and the learning process is guided by a discriminator that can be either classical or quantum. This framework is particularly well-suited for near-term noisy intermediate-scale quantum (NISQ) devices, as it does not require full fault tolerance and can tolerate noise during both state preparation and measurement.

In this work, we explore the foundations of both classical GANs and quantum GANs, and critically assess the potential sources of quantum advantage. We focus on the following guiding research questions: (1) What is a classical GAN, and how does it work? (2) How does the quantum GAN work in the variational quantum generator framework? (3) What could be a source of advantage for a quantum GAN? These questions allow us to analyze not only the conceptual parallels between classical and quantum approaches, but also to highlight the opportunities and challenges that arise when bringing adversarial learning into the quantum domain.

2 Research Objectives and Guiding Questions

We aim to address the following questions based on the framework of variational quantum generators introduced in paper (4):

1. **What is a classical GAN? How does it work?** That is, how do the generator and discriminator interact in a classical Generative Adversarial Network (GAN), and what are the mechanics of their adversarial training?

2. **How does the quantum GAN work?** Specifically, in the variational quantum generator framework, how are classical random variables encoded into quantum states, how is the variational circuit structured, and how are samples generated and used in adversarial training (potentially with both classical and quantum discriminators)?

3. **What could be a source of advantage for a quantum GAN?** In what ways might the quantum implementation—such as encoding via quantum superposition and variational circuits—offer expressive efficiency, compact representation, noise tolerance, or other benefits over its classical counterpart?

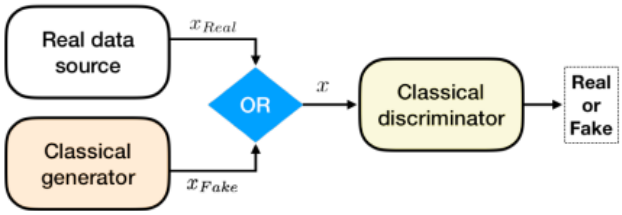

Figure 1: Illustration of a classical GAN: the generator transforms random noise into synthetic data, while the discriminator estimates the likelihood that a sample is real or generated.

3 Findings and Explanations

3.1 Classical GANs: Architecture and Training Mechanism

A generative adversarial network (GAN) is a class of machine learning frameworks and a prominent framework for approaching generative artificial intelligence. The concept was initially developed by Ian Goodfellow and his colleagues in June 2014 [2]. In a GAN, two neural networks, the generator and the discriminator, compete with each other in the form of a zero-sum game, where one agent's gain is another agent's loss.

The core idea of a GAN is based on the "indirect" training through the discriminator, another neural network that can tell how "realistic" the input seems, which itself is also being updated dynamically [7, 2]. This means that the generator is not trained to minimize the distance to a specific image, but rather to fool the discriminator. Figure 1 illustrates the general framework

of GANs. This enables the model to learn in an unsupervised manner, which opens the door to new opportunities, like generating realistic images from noise [2], synthesizing high-resolution data from low-resolution inputs (super-resolution) [3], translating between different domains (for example, turning sketches into photographs) [4, 5], and even creating entirely novel data samples that resemble the training distribution. These capabilities have made GANs a central tool in computer vision and beyond, with applications in image synthesis, style transfer, data augmentation, and creative content generation [2, 4].

Given a prior distribution over the noise parameters $p_z(z)$, the generator consists of a neural network $F_G(z;\Theta_g)$ with parameters Θ_g that generates the distribution p_g. The discriminator is also neural network $F_D(x;\Theta_d)$ that outputs a single scalar corresponding to the probability of x coming from the real data distribution. F_D is trained to maximize the probability of assigning the correct label to both the training examples and the examples coming from F_G. Simultaneously, F_G is trained to minimize $\log(1 - F_D(F_G(z)))$, which is related to the probability of fooling the discriminator. In summary, F_D and F_G play the following adversarial game:

$$\min_G \max_D \mathbb{E}_{x\sim p_{\text{data}}(x)}\big[\log F_D(x)\big] + \mathbb{E}_{z\sim p_z(z)}\big[\log\big(1 - F_D(F_G(z))\big)\big]. \tag{1}$$

Algorithm 1 summarizes the pseudocode for the adversarial learning for classical GANs.

3.2 Quantum GANs: Variational Quantum Generators

QML algorithms use qubits and quantum operations to try to improve the space and time complexity of classical machine learning algortihms [6]. Therefore, attempts to create a Quantum GAN would have appeared sooner or later. Different quantum adaptations of the GAN scheme have published. They can be characterized by the choice of whether the discriminator or the generator will be classical or quantum, the choice of whether the variables are discrete or continuous, and whether the data is classical or quantum [1]. The different variations are depicted in Figure 2. In [1] the authors presented the variational quantum generator (VQG), a framework for quantum generative modeling.

Consider a real data source that outputs observations of an unknown distribution, represented by the variable $x \in \mathbb{R}^N$. The purpose of the variational quantum generator is to produce classical samples x_{Fake} that mimic

Algorithm 1 GAN Training (Minimax Formulation)

Input: dataset $\{x\}$, batch size m, learning rates η_D, η_G, noise prior p_z, generator F_G, discriminator F_D
for epoch $= 1$ **to** N_{epochs} **do**
  for each minibatch $\{x^{(i)}\}_{i=1}^{m}$ **do**
    Sample $\{z^{(i)}\}_{i=1}^{m} \sim p_z(z)$
    Generate fake samples $g^{(i)} \leftarrow F_G(z^{(i)})$
    Compute discriminator loss:

$$L_D = -\frac{1}{m}\sum_{i=1}^{m}\Big(\log F_D(x^{(i)}) + \log(1 - F_D(g^{(i)}))\Big)$$

    Update discriminator: $\phi \leftarrow \phi - \eta_D \nabla_\phi L_D$
    Compute generator loss (minimax form):

$$L_G = \frac{1}{m}\sum_{i=1}^{m}\log\big(1 - F_D(F_G(z^{(i)}))\big)$$

    Update generator: $\theta \leftarrow \theta - \eta_G \nabla_\theta L_G$
  end for
end for

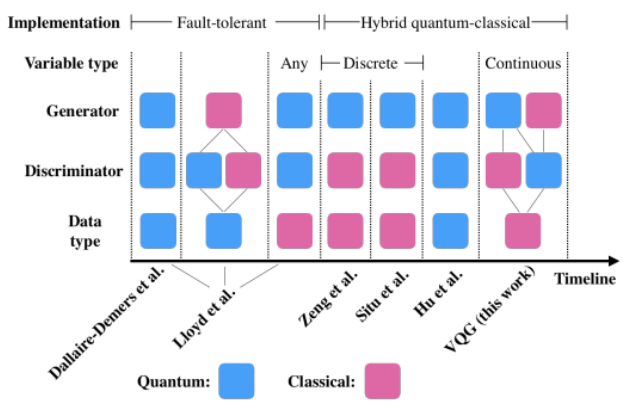

Figure 2: Timeline of the development of quantum generative adversarial network models.

the observed distribution. To achieve this, VQG constructed in the following way, depicted in Figure 3, which includes two variational circuits: a quantum encoding circuit $R(z)$ acting on r qubits, and the generator circuit $G(\Theta_g)$ acting on n qubits with $n \geq r$.

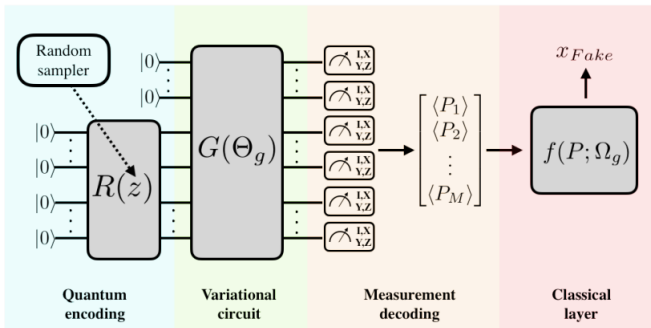

Figure 3: The circuit architecture of VQG comprises a circuit that generates states from a latent space z using the variational circuit $G(\Theta_g)$. The random variable z is mapped to a quantum state via the quantum encoding circuit R. By measuring a fixed set of operators on the generated state, the quantum circuit produces a classical vector

$$P = \big[\langle P_1 \rangle, \ldots, \langle P_M \rangle\big],$$

which then passes through a classical function $f(P;\Omega_g)$ to produce the fake sample x_{Fake}.

The quantum encoding circuit, receives as input a classical random variable $z \sim p_z(z)$ with $z \in \mathbb{R}^O$, and prepares the quantum state

$$R(z)\,|0\rangle^{\otimes r} = |\phi(z)\rangle.$$

This plays the role of the random source in classical GANs, where the space of z corresponds to the latent space in standard generative models. Accordingly, the set of states $\{|\phi(z)\rangle\}$ defines the quantum latent space. Notice that there is two options for choosing how to encode the classical random variables z into quantum states: Amplitude encoding and Variational encoding. **Amplitude encoding:** A classical vector $x \in \mathbb{R}^N$ is normalized and directly mapped into the amplitudes of a quantum state:

$$|x\rangle = \sum_{i=0}^{N-1} x_i |i\rangle.$$

This representation is very compact (only $\log_2 N$ qubits are needed), but preparing such states typically requires deep and complex circuits.

Variational (product) encoding: Each component of a classical input vector is used as a parameter for rotation gates in a parameterized quantum circuit, for example:

$$|z\rangle = \bigotimes_i R_Y(z_i)\,|0\rangle.$$

The authors chose Variational encoding reasoning that works with NISQ hardware (short depth, noise-tolerant), Easier to train (avoids the barren plateau issues associated with large amplitude encoders) and Flexible (can add nonlinearities through extra layers or classical post-processing).

The second variational circuit, $G(\Theta_g)$, functions as the generator model, mapping elements of the latent manifold to the manifold of observable data:

$$G(\Theta_g)\,|\phi(z)\rangle = |\psi(z;\Theta_g)\rangle.$$

Put attention that Θ_g are the trainable parameters.

To extract a classical output from this state, we employ a measurement decoding scheme. Specifically, a sample vector $P \in \mathbb{R}^M$ is obtained by measuring the expectation values of a fixed set of observables, expressed as Pauli strings $\{P_i\}_{i=1}^M$:

$$P = [\langle P_1\rangle, \langle P_2\rangle, \ldots, \langle P_M\rangle],$$

where

$$\langle P_i\rangle = \langle\psi(z;\Theta_g)|\,P_i\,|\psi(z;\Theta_g)\rangle.$$

Finally, the classical vector P is processed by a classical function (which can be represent as NN) $f_g(P;\Omega_g)$ with real parameters Ω_g, producing the synthetic sample

$$x_{\text{Fake}} = f_g(P;\Omega_g).$$

So far, we have described the architecture of the VQG. Now we would like to discuss how to train such a network and what the discriminator structure is.

As in the classical case, the generator and discriminator also play a zero-sum game here, which is expressed by the following cost function:

$$C_d(\Theta_d,\Theta_g) = -\tfrac{1}{2}\,\mathbb{E}_{x\sim p_{\text{data}}(x)}\big[\log F_D(x;\Theta_d)\big] - \tfrac{1}{2}\,\mathbb{E}_{z\sim p_z(z)}\log\Big[1 - F_D\big(F_G(z;\Theta_g,\Omega_g);\Theta_d\big)\Big], \tag{2}$$

And the learning is due to:

$$\min_{\Theta_g} \max_{\Theta_d} C_d(\Theta_d, \Theta_g). \tag{3}$$

Since VQG is mainly about the Generator, we are only interested in the structure of the discriminator because of the calculation of the gradients. Therefore, we can choose whether the discriminator will be classical or quantum. For the classical case, we can choose a standard NN. For the classical case, we can choose a standard neural network. For the quantum case, the authors proposed a quantum circuit constructed as follows: it first encodes the classical sample, then passes it through a variational circuit whose parameters are optimized so that the measured observable corresponds to the probability of the sample originating from the real distribution.

Algorithm 2 summarizes the pseudocode for the adversarial learning of the VQG model.

3.3 Potential Quantum Advantages

A potential source of advantage for a quantum generative adversarial network (QGAN), as outlined in paper (4), lies in the representational power of quantum systems. Classical GANs generate data by transforming random noise through a neural network, but their expressiveness is ultimately limited by the architecture and resources available. In contrast, a QGAN employs a variational quantum circuit to encode classical random variables into quantum states and then manipulate them through unitary operations. Because quantum states naturally exploit superposition and entanglement, the quantum generator can capture correlations and structure within probability distributions in a way that may be exponentially more efficient than classical models.

Another advantage comes from the compactness of quantum descriptions. A small number of qubits can encode information that would otherwise require very large classical vectors. This compact representation allows QGANs to approximate complex continuous distributions with relatively modest resources. Importantly, the authors note that the QGAN framework is well-suited for near-term noisy intermediate-scale quantum (NISQ) devices, since the approach does not require pure quantum states and can tolerate certain levels of noise. This noise-robustness, combined with high expressivity, suggests that QGANs could outperform their classical counterparts in learning and sampling from high-dimensional or classically in-

Algorithm 2 Adversarial learning of a variational quantum generator (VQG).

Result: Optimal Θ_g, Ω_g, and Θ_d
Data: N_s, N_e, S_d, S_g, Initial Θ_g, Ω_g, and Θ_d
for $n = 1$ to N_e **do**
  for $s_1 = 1$ to S_d **do**
    Sample N_s times from $p_z(z)$: $\{z^{(1)}, z^{(2)}, \ldots, z^{(N_s)}\}$
    Sample N_s times from the data distribution: $\{x^{(1)}, x^{(2)}, \ldots, x^{(N_s)}\}$
    Update Θ_d by ascending discriminator's gradient:

$$\nabla_{\Theta_d} \frac{1}{N_s} \sum_i C_d(x^{(i)}, z^{(i)}, \Theta_g, \Omega_g, \Theta_d)$$

  end for
  for $s_2 = 1$ to S_g **do**
    Sample N_s times from $p_z(z)$: $\{z^{(1)}, z^{(2)}, \ldots, z^{(N_s)}\}$
    Update Θ_g and Ω_g by descending generator's gradient:

$$\nabla_{\{\Theta_g, \Omega_g\}} \frac{1}{N_s} \sum_i C_g(z^{(i)}, \Theta_g, \Omega_g, \Theta_d)$$

  end for
end for

tractable distributions, providing a clear potential source of quantum advantage.

3.4 Implementation in code

We have created a code project that implements QGAN in qiskit according to VQG, trains it, and compares the results with Classical GAN. The entire project is in the following GitHub repository: https://github.com/OriMaimon2/QML-Final-Project.git.

From the experiments we got the following observations:
1. Training Losses:

- **Classical GAN** – The losses show oscillations between the generator and the discriminator, indicating a genuine adversarial dynamic.
- **Quantum GAN** – The values remain "stuck": the discriminator around ~ 1.5 and the generator around ~ 0.7, without convergence or oscillatory behavior. This suggests that the generator learns almost nothing and collapses to producing nearly constant outputs.

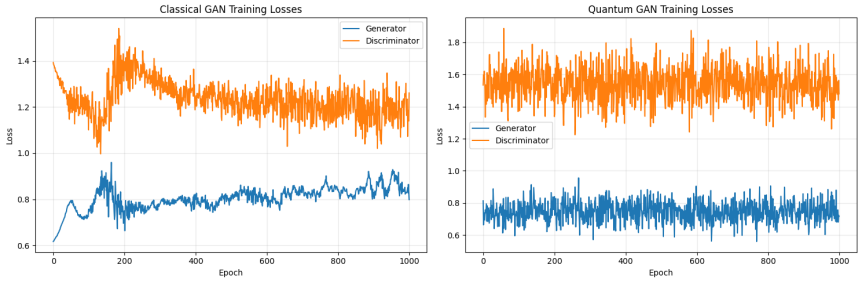

Figure 4

2. Data Distribution

- **Classical GAN** – The histogram shows fixed peaks (e.g., around -1 and 1), meaning the generator locks onto a few discrete points instead of approximating the full data distribution.
- **Quantum GAN** – The output is concentrated around a single value (a spike at 0). This indicates severe mode collapse, where the generator collapses to a single output.

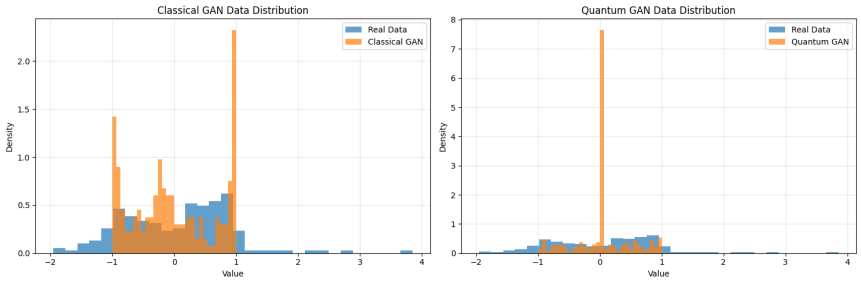

Figure 5

3. Real vs. Generated Samples

- **Classical GAN** – The generated samples (orange points) overlap partially with the real data distribution (blue points) and follow its general shape, although they do not fully capture all the variability of the real distribution. This indicates that the classical generator has learned a reasonable approximation, but with limited expressivity.

- **Quantum GAN** – The generated samples (orange points) collapse along a narrow horizontal line around 0, showing almost no resemblance to the real distribution. This demonstrates mode collapse and a failure of the quantum generator to capture the underlying data distribution.

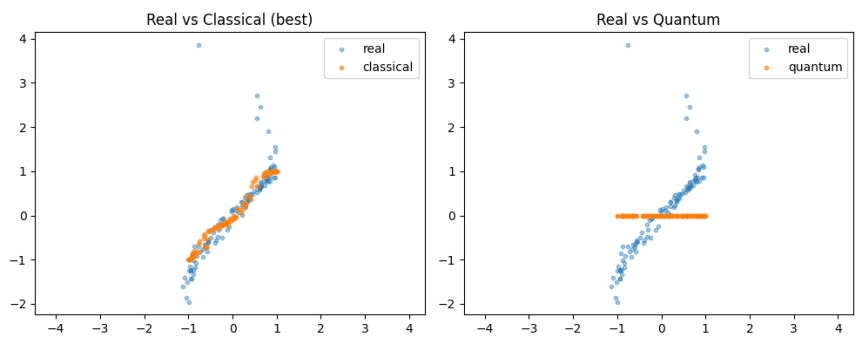

Figure 6

4 Discussion and Future Perspectives

The exploration of quantum generative adversarial networks (QGANs) highlights both the parallels and distinctions with classical GANs. Classical GANs have proven to be highly effective in generating realistic data across many domains, but they remain constrained by the capacity of classical neural networks and the computational resources needed to approximate complex, high-dimensional probability distributions. By contrast, the QGAN framework leverages the representational capacity of quantum systems, which inherently exploit superposition and entanglement. This feature allows for a compact encoding of distributions that may be exponentially difficult to represent classically.

From a practical perspective, the QGAN approach is particularly appealing because it is designed to operate on near-term noisy intermediate-scale quantum (NISQ) devices. Unlike many quantum algorithms that demand fault tolerance and pure quantum states, QGANs can tolerate noise during both the state preparation and measurement stages. This makes them more implementable in realistic quantum hardware environments and offers a pathway for testing genuine quantum advantages in generative modeling.

However, several open challenges remain. The scalability of QGANs is still uncertain, as training large variational quantum circuits may encounter optimization difficulties such as barren plateaus. Additionally, while compactness and expressivity are clear theoretical advantages, demonstrating a concrete performance improvement over state-of-the-art classical GANs remains an open empirical question. Future research should therefore focus on benchmarking QGANs against strong classical baselines and identifying specific application domains—such as quantum chemistry or physics-inspired data—where their advantages may be most pronounced.

In summary, QGANs extend the successful adversarial learning paradigm into the quantum domain, offering potential benefits in representational efficiency, noise tolerance, and compactness. Yet, realizing a tangible quantum advantage will depend on both algorithmic refinements and advances in quantum hardware.

Personal Perspective In our view, the potential quantum gains of QGANs are most likely to appear in domains where classical generative models struggle with representing highly entangled or exponentially large probability distributions, such as quantum chemistry, material discovery, and certain physics-inspired datasets. While we do not expect quantum GANs to immediately surpass classical GANs in general-purpose applications like image

generation, they may provide specialized advantages in problems that are inherently quantum or involve complex correlations that are intractable for classical networks. Considering the current pace of progress in both hardware and hybrid quantum-classical algorithms, I estimate that meaningful, small-scale demonstrations of quantum advantage in QGANs could emerge within the next 5–10 years on noisy intermediate-scale quantum (NISQ) devices. However, achieving broad and practically significant quantum gains will likely require advances in error correction and scalable architectures, which may place large-scale advantages on a longer horizon of 10–20 years.

4.1 Authors contribution

Ori came with the idea of project 4 on quantum GANs, wrote and checked the code. Ariel organized the essay. We both wrote the essay and reviewed the writing together. In addition, we provided feedback on each other's work.

References

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